The 2nd SMPP Workshop

Smart Medicine and Privacy Protection: Challenges and Frontiers in Biomedical and Health Informatics

IEEE BIBM Workshop Proposal for 2026

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Proposal Summary. This proposal requests approval for the 2nd SMPP Workshop at IEEE BIBM 2026. The workshop will provide an interdisciplinary forum on smart medicine, biomedical informatics, and privacy-preserving technologies for healthcare and life science applications.

1. Workshop Title

Smart Medicine and Privacy Protection: Challenges and Frontiers in Biomedical and Health Informatics

2. Introduction to the Workshop

The **1st SMPP Workshop**, held in 2025, attracted broad interest from the research community and received encouraging submissions in the emerging area of smart medicine and privacy protection. It was successfully organized in an online format and provided a valuable forum for interdisciplinary exchange among researchers working at the intersection of biomedical informatics, intelligent healthcare, and privacy-preserving technologies. The strong response to the first edition demonstrated both the **timeliness** of this topic and the **community demand** for a dedicated workshop in this area. Building on that experience, the proposed **2nd SMPP Workshop** aims to further expand this forum and strengthen its role within IEEE BIBM 2026.

The rapid development of biomedical and healthcare information technology is transforming clinical care, personalized medicine, and medical research. At the forefront of this transformation is **intelligent medicine**, characterized by the convergence of artificial intelligence, wearable devices and remote monitoring, the Internet of Things (IoT), 5G-enabled and telemedicine-based healthcare, big data analytics, and health informatics. This trend reflects a deep integration of advanced digital technologies with modern healthcare systems.

Core motivation. While these technologies are driving healthcare toward greater **intelligence**, **precision**, and **efficiency**, they are also reshaping how sensitive **medical**, **clinical**, and **genomic** data are collected, analyzed, and shared across healthcare and biomedical settings. This transformation has raised pressing concerns over **privacy**, **security**, and **governance**, not only as technical issues, but as critical challenges that may affect patient trust, clinical decision-making, cross-institutional medical collaboration, and the responsible advancement of precision medicine.

Representative challenges include:

- **AI-driven smart healthcare:** the use of large volumes of sensitive medical data may introduce risks such as model leakage, data misuse, adversarial attacks, and unauthorized inference, especially under centralized storage and training settings.
- **Wearable devices and remote monitoring:** continuously collected physiological and behavioral data are vulnerable to transmission breaches, device tampering, and identity spoofing.
- **IoT-based healthcare systems:** interconnected sensors and medical devices may be exposed to unauthorized access, data exfiltration, and remote-control attacks, threatening both patient privacy and device integrity.
- **Big data analytics and health informatics:** massive and distributed datasets across institutions and platforms increase the risks of centralized breaches and re-identification through data linkage and correlation.

This workshop aims to serve as an interdisciplinary forum dedicated to **smart medicine** and **privacy protection**. It will focus on the integration of privacy-enhancing technologies, such as federated learning, homomorphic encryption, differential privacy, zero-knowledge proofs, and secure multi-party computation, with smart medical and biomedical applications. The workshop seeks to promote innovative and trustworthy approaches to health data management and to advance research at the intersection of smart medicine, biomedical informatics, and privacy protection.

3. Research Topics Included in the Workshop

The workshop topics include, but are not limited to, the following:

- **Multimodal Data Intelligence and Privacy Challenges in Smart Healthcare**
  - Multimodal medical data collection, analysis, and governance involving wearable devices, medical imaging, electronic health records (EHRs), and remote monitoring systems, especially for high-frequency, strongly correlated, and easily re-identifiable sensitive information streams
  - Semantic sensitivity recognition, patient entity protection, and controllable context generation in clinical information retrieval and knowledge extraction based on natural language processing (NLP)
  - Privacy-preserving mechanisms for AI-assisted decision-making systems in rare disease diagnosis, orphan drug discovery, and drug repurposing, including data anonymization, protection of rare samples, and secure inference
  - Privacy alignment and leakage prevention in the integration of multimodal data (e.g., images, text, physiological signals, speech, and device logs) with high-throughput omics data, particularly in differential analysis and cross-modal fusion scenarios
- **Privacy Protection Mechanisms in Health Information Systems**
  - Fine-grained encryption and access control for EHRs, ICU monitoring data, and clinical records
  - Anonymization and de-identification methods for both structured and unstructured health data
  - Deployment of privacy-enhancing technologies, such as differential privacy and homomorphic

encryption, in real-world hospital information systems, with evaluation of performance, scalability, and usability

- **Privacy-Preserving Artificial Intelligence and Machine Learning**
  - Trade-off optimization between privacy protection and model performance, including privacy-aware neural network design, privacy regularization, model encryption, and pruning strategies
  - Efficient deployment of federated learning in healthcare scenarios, addressing multi-institutional collaboration, data isolation, and model consistency
  - Secure inference and privacy-preserving knowledge distillation techniques for sensitive medical environments
- **Genomic Data Privacy in Precision Medicine**
  - Privacy and security risks in the integration of genomic data across databases, such as single-cell sequencing and clinical phenotype linkage
  - De-identification technologies, revocable consent frameworks, and personalized access control schemes for genomic and precision medicine applications
  - Compliance auditing and privacy assurance mechanisms for genomic research platforms, including issues related to GDPR, IRB approval, and responsible data governance
- **Ethical Governance and Secure Medical Data Sharing**
  - Interoperability standards and secure data exchange protocols across heterogeneous medical systems, such as FHIR and OAuth2.0 + ABAC-based access control
  - Responsibility allocation, ethical review processes, and transparent data disclosure strategies in medical data sharing practices
  - Patient-centered consent negotiation frameworks and data sovereignty mechanisms, including informed data authorization chains and verifiable access logging systems

4. Workshop Style

The proposed workshop will be held in a **fully online** format. The workshop sessions will be conducted through a virtual meeting platform to support paper presentations, keynote talks, and interactive discussions. The specific platform and detailed logistical arrangements will be finalized later and announced to the authors upon paper acceptance.

5. Important Dates

- **Call for Papers:** June 25, 2026
- **Full Paper Submission Deadline:** September 27, 2026
- **Notification of Acceptance:** October 18, 2026
- **Camera-Ready Submission:** November 8, 2026
- **Workshop Date:** December 1–4, 2026

6. Program Chairs / Co-Chairs

The organizing team includes chairs from Huazhong University of Science and Technology, China, and the NUS School of Computing, National University of Singapore, Singapore. The workshop organization is supported by the Hubei Key Laboratory of Distributed System Security and the Hubei Engineering Research Center on Big Data Security. Prof. Songfeng Lu serves as Director and is also affiliated with the Shenzhen Huazhong University of Science and Technology Research Institute.

Prof. Songfeng Lu	Director, School of Cyber Science and Engineering, Huazhong University of Science and Technology, China. *Expert in bioinformatics and data privacy*. Email: lusongfeng@hust.edu.cn.
Dr. Zhi Lu	Researcher Fellow, NUS School of Computing, National University of Singapore, Singapore. *Expert in privacy computing and genomic data security*. Email: luzhi@ieee.org.
Dr. Renfei Shen	School of Cyber Science and Engineering, Huazhong University of Science and Technology, China. *Expert in medical data sharing and ethical governance*. Email: shenrenfei@hust.edu.cn.

7. Program Committee Members

The Program Committee includes experts from multiple institutions with complementary backgrounds in smart medicine, biomedical informatics, privacy protection, medical systems, and artificial intelligence. Assoc. Prof. Hewang Nie is with the School of Computer Science and Engineering, School of Software, and School of Artificial Intelligence, Guangxi Normal University, China. Dr. Zilong Wang, Dr. Shuai Guo, Dr. Junming Li, and Dr. Yutong Wu are with the School of Cyber Science and Engineering, Huazhong University of Science and Technology, China. Dr. Huan Liu is with the Hubei Engineering Research Center on Big Data Security, the Hubei Key Laboratory of Distributed System Security, the National Engineering Research Center for Big Data Technology and System, the Services Computing Technology and System Lab, the Cluster and Grid Computing Lab, and the School of Cyber Science and Engineering, Huazhong University of Science and Technology, China.

- **Assoc. Prof. Hewang Nie**, Expert in medical model watermarking and copyright protection
- **Dr. Zilong Wang**, Expert in privacy-preserving AI and machine learning
- **Dr. Shuai Guo**, Expert in innovation at the intersection of medicine and technology
- **Dr. Junming Li**, Expert in genomic data privacy in precision medicine
- **Dr. Yutong Wu**, Expert in bioinformatics and AI privacy
- **Dr. Huan Liu**, Expert in medical system vulnerability detection and fuzz testing

8. Invited Keynote Speaker

- **Assoc. Prof. Hewang Nie**, Expert in medical model watermarking and copyright protection